UCC 103 PRINCIPLES OF COMPUTING
CHAPTER SIX: INTRODUCTION TO COMPUTER NETWORKS

6.1 Definition:

- ✓ A computer network is a group of computer systems and other computing hardware devices that are linked together through communication channels to facilitate communication and resource-sharing among a wide range of users (Technopedia).
- ✓ A computer network is a set of computers connected together for the purpose of sharing resources.

The most common resource shared today is connection to the Internet. Other shared resources can include a printer or a file server. The Internet itself can be considered a computer network.

NB: Connection is physically established through cables, lasers, microwaves, fiber optics and communication satellite as well as Virtually through WiFi (wireless networking technology that uses radio waves)

6.2. Objectives of Computer Networks

The following are the objectives of the computer networks.

1) Resource sharing is the main objective of the computer network. The goal is to provide all the program, date and hardware is available to everyone on the network without regard to the physical location of the resource and the users.
2) High Reliability computer files: It is achieved by replicating the files on two or more machines, so in case of unavailability (due to fail of Hardware) the other copies can be used.
3) Cost Saving: Computer organization has helped organization in saving money. This is due to the fact that the small computer has much better price to the performance ratio comparison than the large computer like mainframe. Mainframe computer are approximately ten times faster than the microcomputers, but they cost thousands times more. As a result of this imbalance, organization has preferred to install interconnected microcomputer connected to the mainframe computer.
4) Increase Performance: Computer network have provided means to increase system performance as the work load increases (load balancing
5) Remote / Virtual Working: Computer network help people who live or work apart to report together. So, when one user prepared some documentation, he can make the document online enabling other to read and convey their opinions. Thus computer network is a powerful communication medium.
6) Enhance Computer System Security: Only authorized user can access resource in a computer network. Users are authenticated by their user name and password.

Hence it is not possible to access the data without proper account. This increases security.

7) Load Sharing: Both the program and data are transmitted to a remote computer to equalize the load between the two facilities.
8) Message Service: A network can be used to handle interpersonal message transmissions. This type of service can also be used for educational services and conference activities. However, it is not an important motivation for a network of scientific computers.
9) Data Sharing: The program is sent to a remote computer where a large data base exists. This type of operation will be particularly useful where data files are too large to be duplicated economically
10) Program Sharing
11) Remote Service

6.3. Characteristics / Properties of Computer networks:

The Characteristics of computer networks are;

1) Share network and computing resources: In a networked environment, each computer on a network may access and use resources provided by devices on the network, such as printing a document on a shared network printer.
   - ✓ Distributed computing uses computing resources across a network to accomplish tasks.
   - ✓ Connect a printer, scanner, or a fax machine to one computer within the network and let other computers of the network use the machines available over the network.
2) Permit sharing of files, data, and other types of information: In a network environment, authorized users may access data and information stored on other computers on the network. The capability of providing access to data and information on shared storage devices is an important feature of many networks.
3) Facilitate communications: Using a network, people can communicate efficiently and easily via email, instant messaging, chat rooms, telephone, video telephone calls, and video conferencing.
4) Reliability makes easy to use an alternative source for data communication in case of hardware failure or connectivity issues
5) Performance/ Speed by measuring the speed of data transmission with number of users, connectivity and the software used
6) Scalability increases the system performance by adding more processors.
7) Security is the main characteristics of Computer network where you can take necessary steps for protecting your data from unauthorized access
8) Availability. Availability is typically measured in a percentage based on the number of minutes that exist in a year. Therefore, uptime would be the number of minutes the network is available divided by the number of minutes in a year.
9) May be insecure: A computer network may be used by computer hackers to deploy computer viruses or computer worms on devices connected to the

network, or to prevent these devices from normally accessing the network (denial of service).

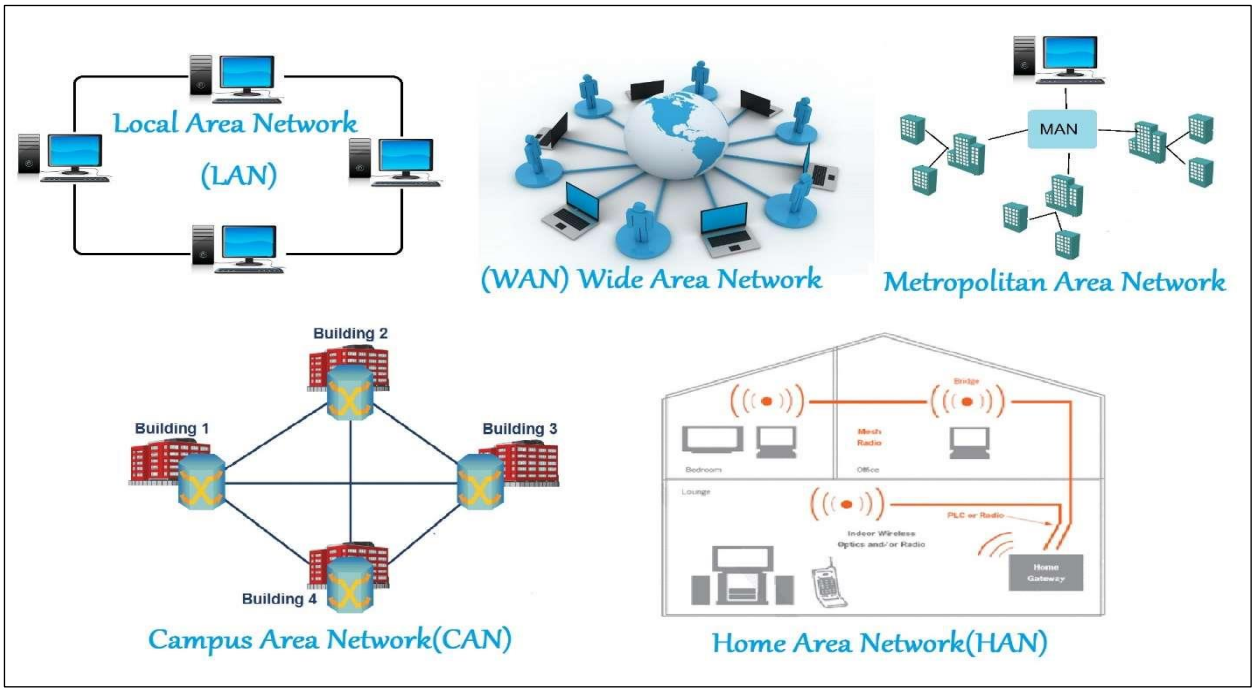
10) May interfere with other technologies: Power line communication strongly disturbs certain forms of radio communication, e.g., amateur radio. It may also interfere with last mile access technologies such as ADSL and VDSL.
11) May be difficult to set up: A complex computer network may be difficult to set up. It may also be very costly to set up an effective computer network in a large organization or company.
12) Has a Cost: includes the cost of the network components, their installation, and their ongoing maintenance.
13) Topology: describes the physical cabling layout and the logical way data moves between components.

6.4. Type of Networks / Network Terminologies

There are many types of computer networks, including the following

1) **Intranet:** An intranet is a set of networks, using the Internet Protocol and IP-based tools such as web browsers and file transfer applications, that is under the control of a single administrative entity. That administrative entity closes the intranet to all but specific, authorized users. Most commonly, an intranet is the internal network of an organization. A large intranet will typically have at least one web server to provide users with organizational information.
2) **Extranet:** An extranet is a network that is limited in scope to a single organization or entity and also has limited connections to the networks of one or more other usually, but not necessarily, trusted organizations or entities—a company's customers may be given access to some part of its intranet—while at the same time the customers may not be considered *trusted* from a security standpoint.
3) **Local area network**: A local area network (LAN) is a network that connects computers and devices in a limited geographical area such as home, school, computer laboratory, office building, or closely positioned group of buildings. Each computer or device on the network is a node.
4) **Wide area network**: A wide area network (WAN) is a computer network that covers a large geographic area such as a city, country, or spans even intercontinental distances, using a communications channel that combines many types of media such as telephone lines, cables, and air waves. A WAN often uses transmission facilities provided by common carriers, such as telephone companies.
5) **Enterprise private network**: An enterprise private network is a network built by an enterprise to interconnect various company sites, e.g., production sites, head offices, remote offices, shops, in order to share computer resources.
6) **Virtual private network**: A virtual private network (VPN) is a computer network in which some of the links between nodes are carried by open connections or virtual circuits in some larger network (e.g., the Internet) instead of by physical wires.

7) CAN or Campus Area Network: – It is a Computer network designed to be used by a particular campus or a military base with Computers placed within a limited area.
8) HAN or Home Area Network: – The network is connected within user home that connects various digital devices over a network for communicating with other devices.

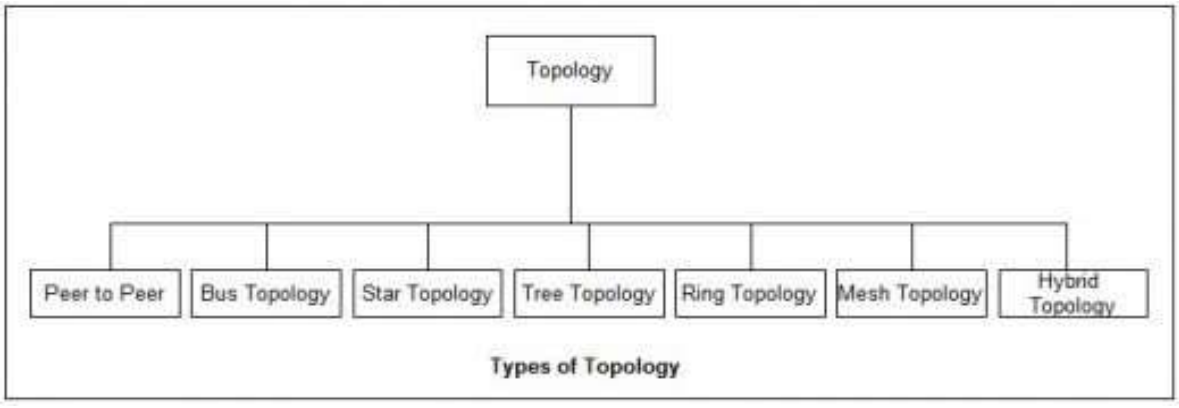


6.5. Computer Network Topology

The graphical arrangement of computer systems, or nodes to form a computer network, is called the network topology. A network topology can be one of two types;

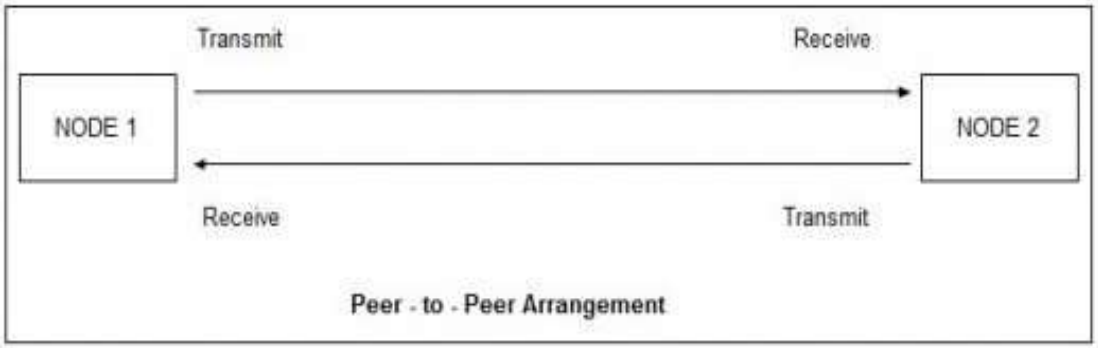
1) Physical topology: physical arrangement of computer nodes based on hardware configuration
2) Logical topology; - defines how data is flowing through the network.

6.6. Types of Computer Network Topologies



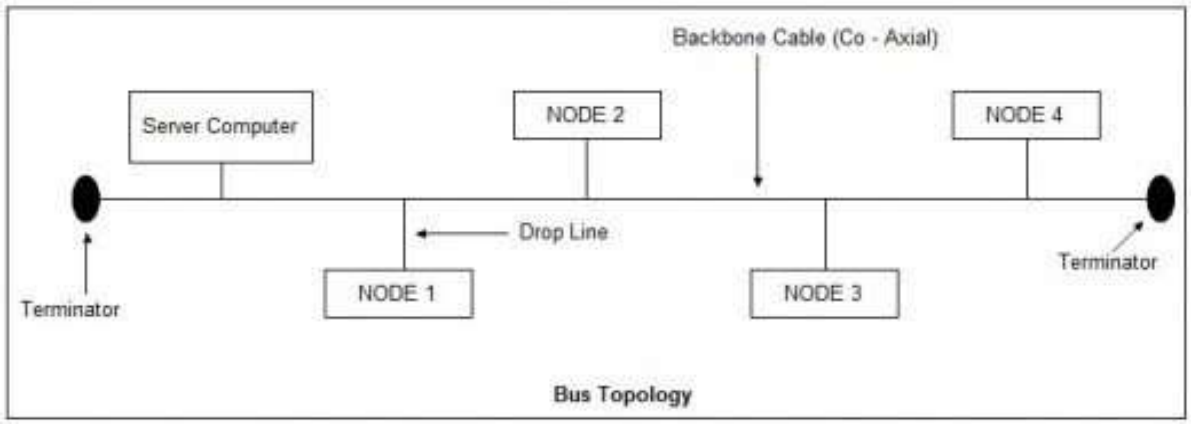
6.6.1. Peer–to–Peer Arrangement

Sometimes this is called as a point-to-point link. Each point–to–point link contains one transmitter and one receiver. Each station receives data exactly from one transmitter and each transmitter transmits data to exactly one receiver. Receiving and transmission process can be done over a single wire or can use separate wires for better performance.



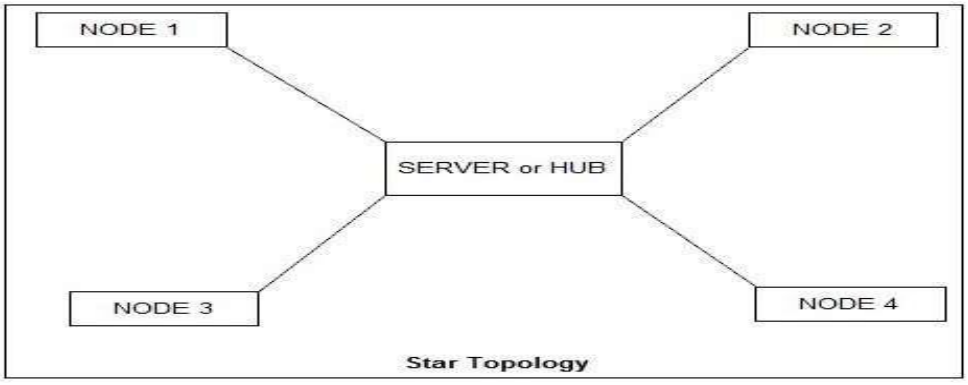
6.6.2. Bus Topology

Bus topology is also called as point–to–multi–point arrangement. In a bus topology, all nodes or devices are linked with one transmitter or server computer via a single cable (mostly coaxial cable) called Backbone. All nodes are connected to the bus cable by drop lines. A drop line is a connection running between the nodes and the main cable.



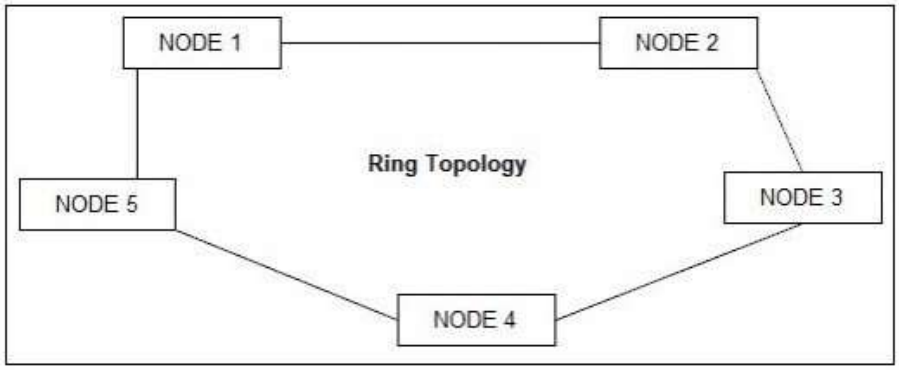
6.6.3. Star Topology

In a star topology, each node has a dedicated point–to–point connection only with a central server, normally called a **hub.** The nodes are not directly connected to each other. Hence, direct data flow is not allowed in between the nodes. The central server or hub acts as a data exchanger as one node sends data exactly to hub and hub transmits this data to another node.



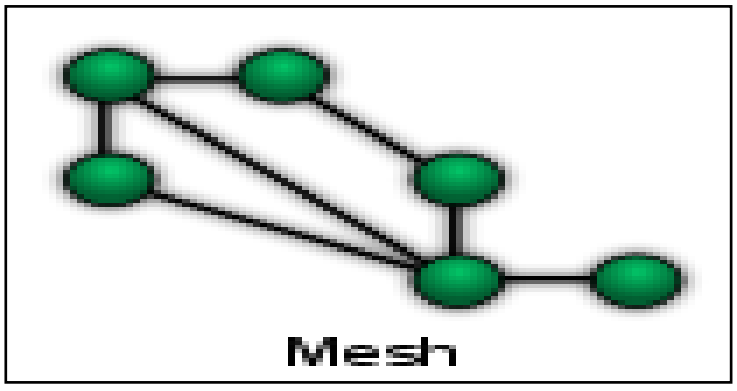
6.6.4. Ring Topology

In a ring topology, each node has a dedicated point–to–point line configuration only with the two nodes on either side of it. A signal is passed along the ring in one direction, from node to node, until it reaches its receiver. Each node in the ring is integrated as a repeater. When a node receives a signal intended from another node, its repeater regenerates the bits and passes them along.



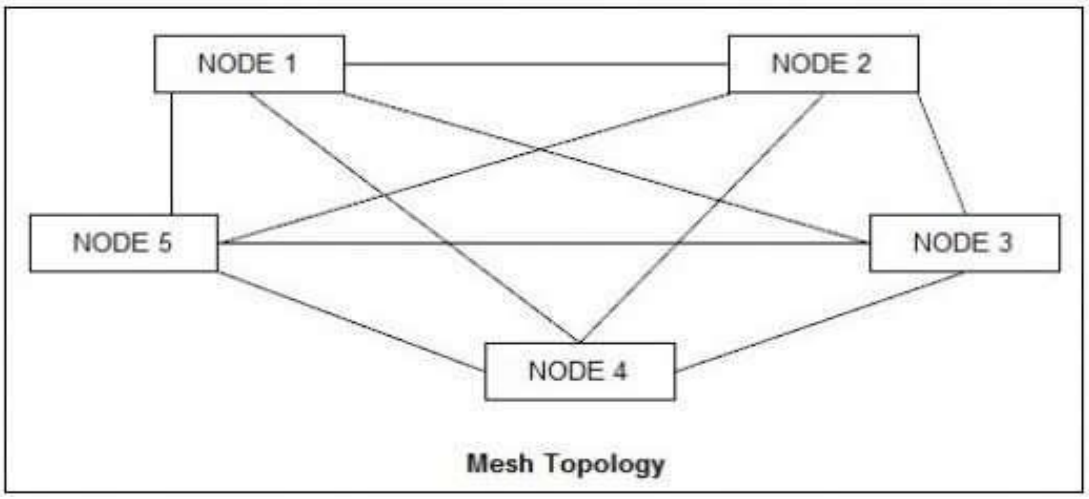
6.6.5. Mesh Topology

In a mesh topology, each node has a dedicated point–to–point connection with an arbitrary number of neighboring nodes in the network in such a way that there is at least one traversal from any node to any other. Dedicated connection means data can only be flown between two nodes, it connects.



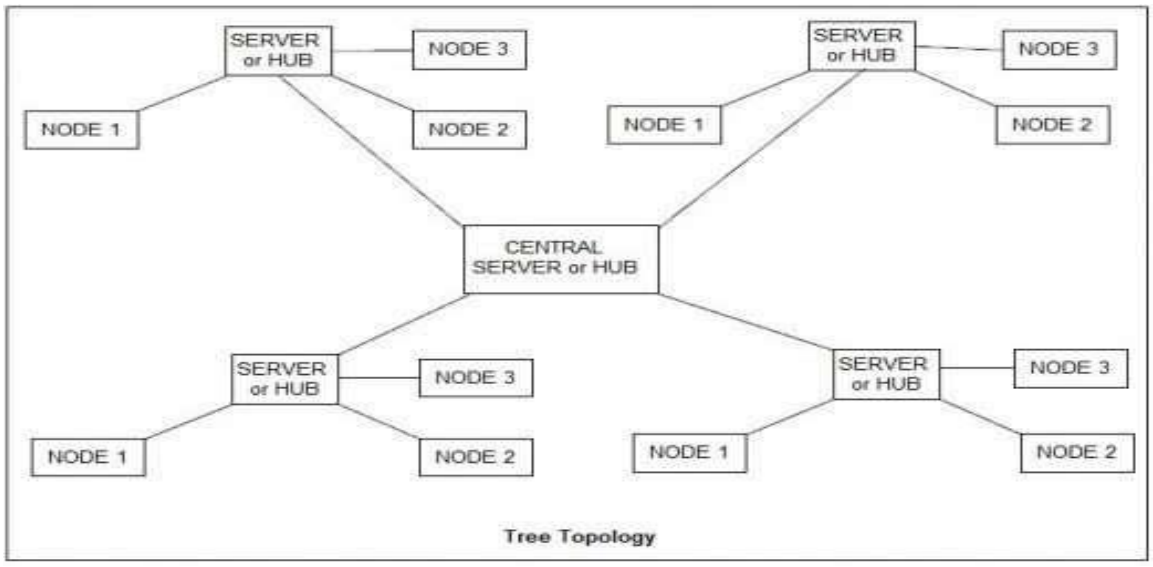
6.6.6. Fully Connected

In a fully connected topology, each node has a dedicated point–to–point connection with every other node in the network



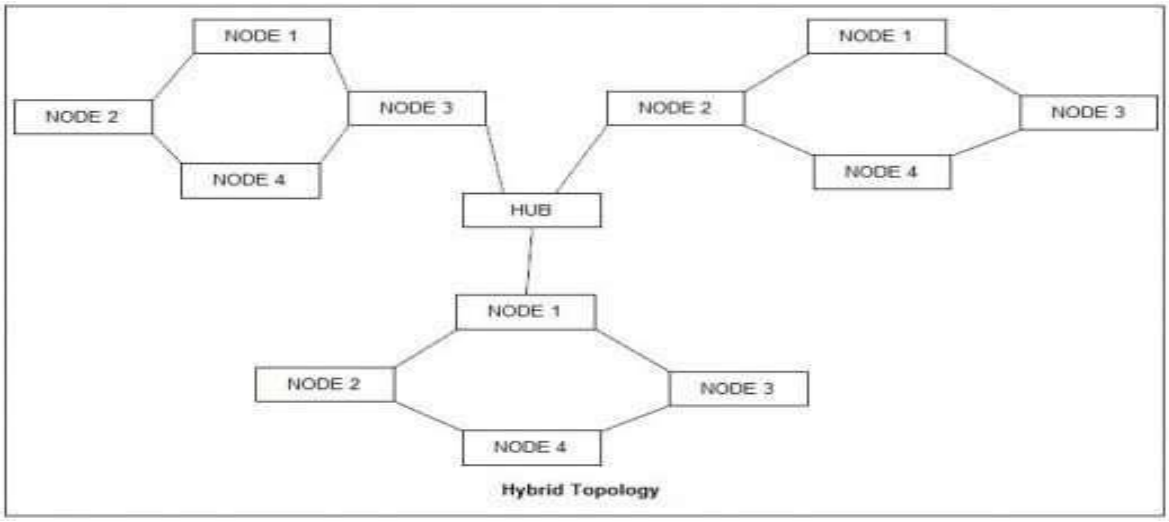
6.6.7. Tree Topology

Tree topology is a combination of one or more star topology arrangements. All the sub- central hubs are connected to a main central hub to form a tree topology. A cable TV network is a good example of this type of topology.



6.6.8. Hybrid Topology

This topology is a common arrangement of one or more topologies described above. It means one or more above described topologies connected with each other to form a hybrid network arrangement.



Star - Ring - Hybrid Topology

6.7. Application of Computer Networks:

Some of the network applications of the different fields are the following.

1) Marketing and sales: Marketing professional uses them to collect exchange and analyze data relating to customer needs and product development cycles.
2) Sales application includes Teleshopping, which uses order entry computers or telephone connected to an order processing network, and online reservation services for railways, hotels, airlines, restaurants theatre etc.
3) Financial services: It include credit history searches, foreign exchange and investment services and electronic fund transfer (EFT), which allow a user to transfer money without going to bank.
4) Manufacturing: computer networks are used today in many aspects of manufacturing, including the manufacturing processes itself. Two aspects that uses network to provide essential services are computer Assisted design (CAD) and computer Assisted manufacturing (CAM), both of which allow multiple user to work on a project simultaneously.
5) Electronic Messaging: E-mails transfer the messages between two and more users in a network. With this application user can transfer the information in the form of text, picture and voice.
6) Directory Services: It allows list of files to be stored in central location to speed up the worldwide search operation. E.g. search engines like Google, Britannia, and Yahoo etc.
7) Information Services: It includes Bulletin Boards and data bank. A 'www' site offering the technical specification for a new product in a information services.
8) Electronic Data Exchange (EDI): EDI allows business information (including documents such as purchase orders and services) to be transferred without using paper

9) Teleconferencing: It allows conference to occur without the participant being in the same location. It includes:
    - ✓ Text Conferencing: Participant communicates through their keywords and computer monitors.
    - ✓ Voice Conferencing: Participant at a number of locations communicates simultaneously through phone (talk).
    - ✓ Video Conferencing: Participant can see as well as talk to another.
10) Cellular Telephone: Wireless phone communication even while travelling through long distance.
11) Cable TV: This is widest usable thing today throughout the world.

6.8. Computer Networks Components

The network is made up of the following components:

1) Hardware components
    - ✓ hosts (PCs, laptops, handhelds)
    - ✓ Electronic Devices
2) Communication line and devices routers & switches
3) Software Components/Protocols
   TCP/IP Software is the intelligence that causes all of the components to function together. The most popular network software today uses what is known as the TCP/IP protocol suite, or stack. The suite is constructed from actual layers of software, where each has its own function.
4) Applications (network services)
5) Humans and service agents

6.8.1. Hardware Components

A part from the computer, other hardware components of a computer network includes;

1) **Network Adapters Modems/Interface: A** modem is a device that converts computer information into electronic signals that can be transmitted over a phone line. At the other end of the line, another modem converts the telephone signals back into computer data. There are different types of network adapters that are used to configure or setup the internet technology on your operating system.
2) **Switch:** A switch is a telecommunication device which receives a message from any device connected to it and then transmits the message only to that device for which the message was meant. A device that allocates, filters and forwards traffic (data gram) between ports based on the data gram address.
3) **Router:** A router is a device that forwards data packets between computer networks. A router is connected to two or more data lines from different networks. When a data packet comes in on one of the lines, the router reads the address information in the packet to determine its ultimate destination.

4) **Gateway:** device sitting at a network node for interfacing with another network that uses different protocols. A gateway acts as a protocol translators, thus allowing interconnections of networks with different network protocol technologies
5) **Firewalls:** is a software or hardware-based network security system that controls the incoming and outgoing network traffic by analyzing the data packets and determining whether they should be allowed through or not, based on applied rule set. Firewalls can be defined in many ways according to your level of understanding. A firewall establishes a barrier between a trusted, secure internal network and another network (e.g., the Internet) that is not assumed to be secure and trusted.
6) Bridge: a device that connects multiple network segments along the data link layer. Bridges broadcast to all ports except the port on which the broadcast was received. Bridges come in three basic types:
    - ✓ Local bridges: Directly connect LANs
    - ✓ Remote bridges: Can be used to create a wide area network (WAN) link between LANs. Remote bridges, where the connecting link is slower than the end networks, largely have been replaced with routers.
    - ✓ Wireless bridges: Can be used to join LANs or connect remote stations to LANs.
7) Repeater/Hub: device to amplify or regenerate digital signals received while sending them from one part of a network into another. A repeater is an electronic device that receives a signal, cleans it of unnecessary noise, regenerates it, and retransmits it at a higher power level, or to the other side of an obstruction, so that the signal can cover longer distances without degradation.
8) Media (Cabling) Typical network cable connector of course, none of these networking devices work well unless they are connected to each other, and that is done with various media. The most ubiquitous media is commonly called Ethernet cabling, which is actually one of several categories of unshielded twisted pair (UTP) wiring.